Paper review · Visionary

When should a fine-tuned OpenVLA ask for help?

A follow-up research project and an application for OpenVLA: An Open-Source Vision-Language-Action Model. Kim, Pertsch, Karamcheti et al., Stanford, UC Berkeley, Toyota Research Institute, Google DeepMind, Physical Intelligence and MIT.

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bushuyeu.com/review-openvla-visionary

What OpenVLA makes possible

+16.5 points

absolute success over the closed 55B RT-2-X across 29 tasks on two robots, with 7× fewer parameters.

1.4% of the weights

LoRA at rank 32 matches full fine-tuning (68.2% vs 69.7%), in 10–15 hours on one A100.

4-bit

inference matches bfloat16 in real tasks with less than half the memory.

256 bins per dimension

every action dimension is one token, so every action comes with a probability.

7B: SigLIP and DINOv2 into Llama 2 (Prismatic-7B); 970k Open X-Embodiment episodes; 21,500 A100-hours. Limits it names: one image, ~6 Hz on an RTX 4090, reliability below 90%.

Proposed research

Does cheap fine-tuning break a VLA's confidence?

Confidence drives help requests

Recent monitors turn token-based VLAs' action confidence into help requests (INSIGHT, ReconVLA, VLAConf). They assume it is calibrated: 90% sure means about 90% success.

LoRA makes language models overconfident

Fine-tuning on small datasets does, and Bayesian LoRA repairs much of it (Laplace-LoRA, LoRA ensembles). For robot policies this is untested: the one VLA calibration study measured released checkpoints, all LoRA at rank 32.

Hypotheses

H1 OpenVLA becomes more overconfident the fewer demonstrations it is fine-tuned on, and LoRA more than full fine-tuning at equal success.

H2 Bayesian LoRA lowers calibration error more than post-hoc fixes (temperature, action-wise Platt scaling) and keeps success the same.

H3 The gap grows under LIBERO-Plus camera, lighting and layout changes, where a robot most needs to ask for help.

If H1 holds, every confidence monitor on a LoRA-tuned VLA inherits the error, and the fix is cheap. If not, post-hoc calibration on a small labelled set is enough.

Design

Model	OpenVLA-7B from its released pretrained weights.
Fine-tuning	Full; LoRA rank 8; LoRA rank 32 (the default).
Data	10% and 100% of each LIBERO suite's demonstrations.
Fixes	None; temperature; action-wise Platt; Laplace-LoRA; 4 LoRA adapters as an ensemble.
Tasks	LIBERO Spatial, Object, Goal, Long; LIBERO-Plus perturbations for H3.
Metrics	Success; calibration error, Brier score and failure-prediction AUROC of episode-success confidence.
Trials	500 per configuration (10 tasks × 50).

Compute and risks

Compute, estimated

LoRA: 10–15 A100-hours per run, ~60 GB at batch 16. A first pass on two suites is 18 runs: roughly 180–270 A100-hours.

Evaluation: ~6 actions per second, episodes capped at 220–520 steps: at most ~5 GPU-hours per configuration on Spatial, 12 on Long.

Risks

Token-based success confidence is a proxy; the calibration study found it weakest late in failed episodes.

OpenVLA still fails about a quarter of LIBERO episodes (76.5% average): enough failures to calibrate against. An ensemble costs 4× the inference.

Rehearse in a digital twin before the robot works

1. **Scan** a few minutes of phone video of the workspace and its objects.
2. **Rebuild** it as a reconstruction with object meshes and the robot in a simulator.
3. **Rehearse** each candidate checkpoint 200–300 times on the target tasks.
4. **Calibrate** confidence and choose the room's ask-for-help threshold.
5. **Deploy and check** whether the real robot's caught failures match the twin's prediction.

What a user gets, and what it builds on

Before the first real trial

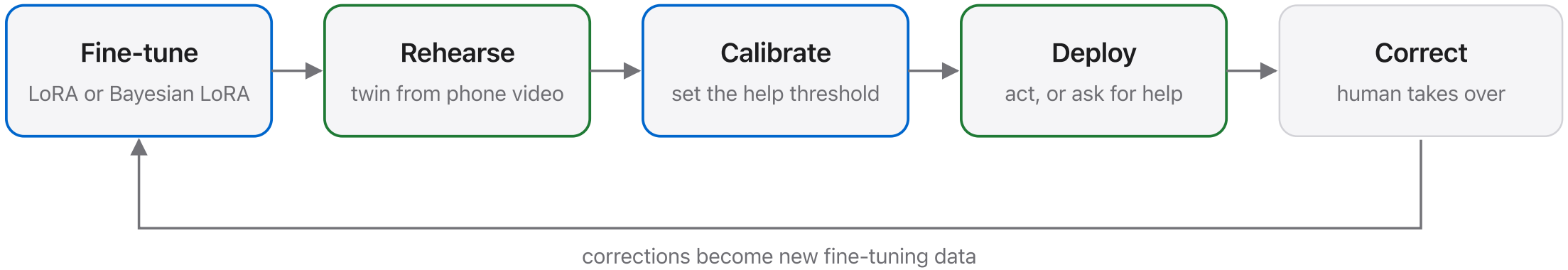
- Which fine-tuned checkpoint to run in this room.
- The failures to expect, as rollout videos.
- The confidence threshold for asking for help.

Existing pieces

SIMPLER and ReVeal: simulated and reconstructed success tracks real success. RialTo: twins from quick scans. This routine joins them and adds the calibration step.

Risk: lower reconstruction fidelity weakens sim-to-real agreement (ReVeal), so the transfer is what gets measured.

How the two fit together



■ research project: fine-tuning and calibration

■ application: twin and deployment

What this builds on

Work	What it does	What this proposal adds
Zollo and Zemel, 2025	Calibration of released fine-tuned VLAs on LIBERO; Platt scaling.	Varying the fine-tuning.
INSIGHT, RECALL	Help triggers and recovery data from token uncertainty.	Calibration of that uncertainty.
ReconVLA, VLAConf	Conformal and representation-level confidence.	Labelled rollouts from a twin.
Laplace-LoRA, LoRA ensembles	Fix overconfidence of fine-tuned LLMs.	A test on a robot policy.
SIMPLER, ReVeal, RialTo	Sim and twin evaluation; twins from scans.	Calibration per room.

Timeline

Plan for one semester

Weeks 1–3	**Reproduce the baseline** Calibration of the released OpenVLA LIBERO checkpoints.
Weeks 4–8	**Fine-tuning grid (H1, H2)** Full and LoRA, two data sizes, Bayesian LoRA and post-hoc fixes; Spatial and Long.
Weeks 9–10	**Shift (H3)** The same checkpoints under LIBERO-Plus perturbations.
Weeks 11–14	**One room, end to end** Phone-video twin, rehearsal, calibration, real-robot check.
Week 15	**Write-up** H1–H3 and the twin-to-real transfer of calibration.

Sources

1. Kim et al. OpenVLA, arXiv:2406.09246: §3.5, §5.3 and Table 1, §5.4, §6, Appendix E (LIBERO).
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6. Liu et al. LIBERO, 2023; LIBERO-Plus, 2025.
7. Li et al. SIMPLER, 2024; Wang et al. ReVeal, 2026.
8. Torne et al. RialTo, 2024.
9. This page: bushuyeu.com/review-openvla-visionary.